

Program 4

Objective: Create a Dockerized Flask application and push it to GitHub using Personal Access Token (PAT).

Project Structure

```
lab4/  
|  
├─ app.py  
├─ requirements.txt  
├─ Dockerfile  
├─ README.md  
└─ lab-4.png
```

Source Code

Dockerfile

```
FROM python:latest  
WORKDIR /app  
COPY requirements.txt .  
RUN pip install --no-cache-dir -r requirements.txt  
COPY . .  
EXPOSE 5000  
CMD ["python", "app.py"]
```

app.py

```
from flask import Flask  
  
app = Flask(__name__)  
  
@app.route('/')  
def hello():  
    return "hello world from container"
```

```
if __name__ == '__main__':  
    app.run(host="0.0.0.0", port=5000)
```

requirements.txt

```
Flask==2.3.3
```

Setup and Execution

Build Docker Image

```
1rv24mc078_prateek_kumar@arch ~/d/lab-4 sudo docker build -t program4 .
```

Run Docker Container

```
1rv24mc078_prateek_kumar@arch ~/d/lab-4 sudo docker run -p 5000:5000  
program4
```

Open in Browser

```
http://localhost:5000
```

GitHub Personal Access Token (PAT)

Initialize Git Repository

```
1rv24mc078_prateek_kumar@arch ~/d/lab-4 git init  
1rv24mc078_prateek_kumar@arch ~/d/lab-4 git branch -M main  
1rv24mc078_prateek_kumar@arch ~/d/lab-4 git add .  
1rv24mc078_prateek_kumar@arch ~/d/lab-4 git commit -m "Initial commit"
```

Push to GitHub

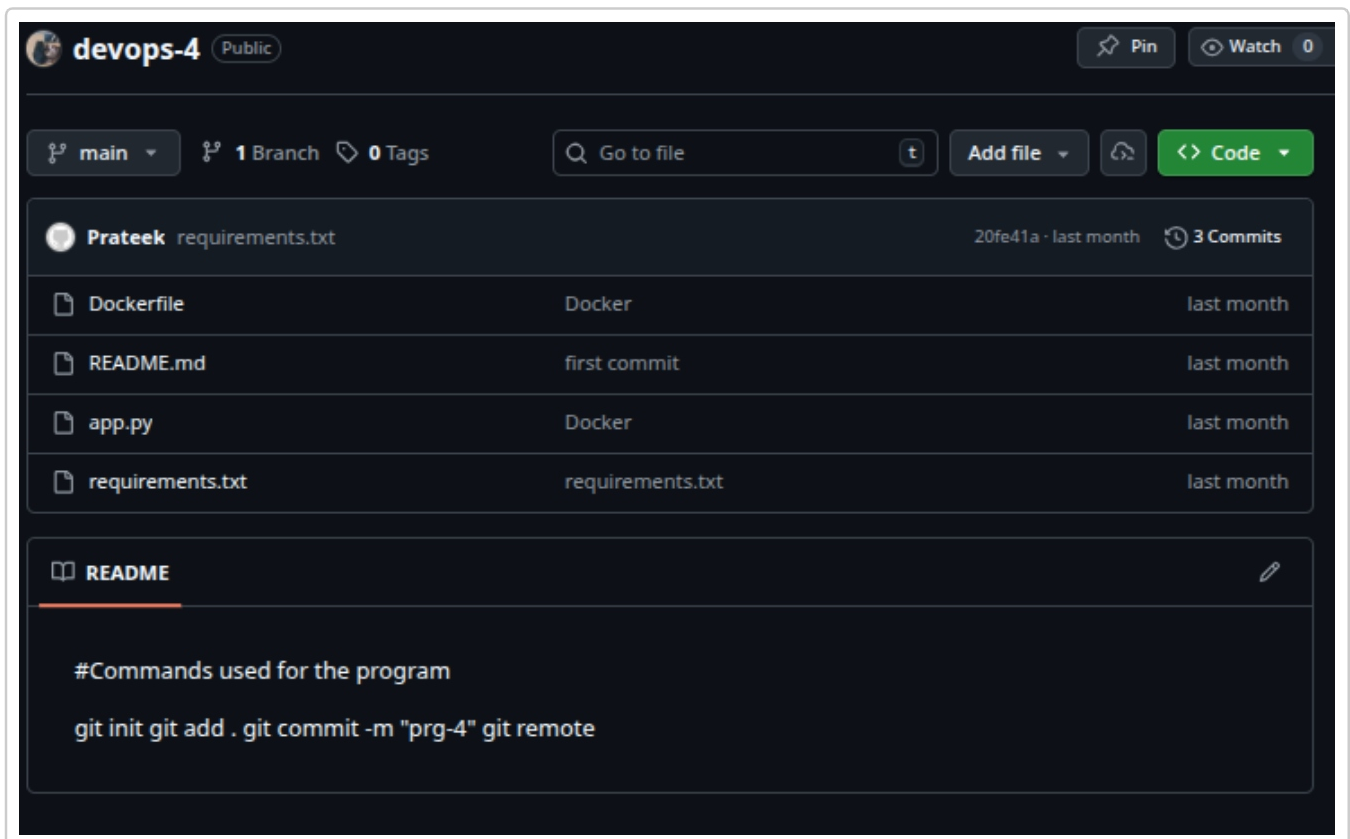
```
1rv24mc078_prateek_kumar@arch ~/d/Lab-4 git remote add origin
https://github.com/<username>/<repo-name>.git
1rv24mc078_prateek_kumar@arch ~/d/Lab-4 git push -u origin main
```

When prompted:

- **Username:** GitHub username
- **Password:** Paste generated PAT

Output Screenshot

Flask Application Output in Browser:



Result

The Flask application runs successfully inside a Docker container and is pushed to GitHub.